OL7014 Integer Programming

Homework 2 — Solutions

Due Oct 13, start of class

Problem 1. Cadets and a married couple **(20 points)**

Three cadets (A, B, C) are commissioned into three branches, one each, to maximize the total fit:

fit	Signal	Artillery	Engineers
cadet A	2	5	1
cadet B	5	3	2
cadet C	2	3	5

(a) **(5 pts)** Formulate the IP and write it as max $c^\top x$ s.t. $Ax = b$, $x \in \{0,1\}^9$, with x flattened row by row.

Solution. $x_{ij} = 1$ iff cadet i goes to branch j; $x = (x_{A,\text{Sig}}, x_{A,\text{Art}}, x_{A,\text{Eng}}, x_{B,\text{Sig}}, \ldots, x_{C,\text{Eng}})$. Rows 1–3 say each cadet gets one branch, rows 4–6 say each branch gets one cadet.

$$c = \begin{pmatrix} 2 & 5 & 1 & 5 & 3 & 2 & 2 & 3 & 5 \end{pmatrix}^\top, \qquad A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

(b) **(4 pts)** Show that A is TU with the checklist (give M_1, M_2). What does this say about the LP relaxation?

Solution. (i) every entry is 0 or 1; (ii) every column has exactly two nonzeros — column (i,j) is $e_i + e_{3+j}$; (iii) take $M_1 = \{1,2,3\}$ (the cadet rows) and $M_2 = \{4,5,6\}$ (the branch rows): each column has one 1 in each block, so both sums are 1. All three hold, so A is TU: it is the incidence matrix of the bipartite graph cadets – branches.

With $b = \mathbf{1}$ integral, Hoffman–Kruskal gives integral corners, so the LP relaxation has an integral optimum.

(c) **(3 pts)** Solve the LP relaxation with `gurobipy` and confirm the solution is integral.

Solution.

```
import gurobipy as gp
from gurobipy import GRB

fit = [[2, 5, 1], [5, 3, 2], [2, 3, 5]]        # rows A,B,C   cols Sig,Art,Eng
R = range(3)
m = gp.Model("cadets")
x = m.addVars(R, R, lb=0, ub=1, name="x")      # CONTINUOUS: the relaxation
m.setObjective(gp.quicksum(fit[i][j]*x[i, j] for i in R for j in R), GRB.MAXIMIZE)
m.addConstrs(x.sum(i, "*") == 1 for i in R)    # each cadet once
m.addConstrs(x.sum("*", j) == 1 for j in R)    # each branch once
m.optimize()
print("z =", m.ObjVal, " x =", {(i, j): x[i, j].X for i in R for j in R if x[i, j].X > 1e-6})
```

```
z = 15.0  x = {(0, 1): 1.0, (1, 0): 1.0, (2, 2): 1.0}
```

Integral with no integrality declared: A → Artillery, B → Signal, C → Engineers, $z = 5 + 5 + 5 = 15$.

(d) **(3 pts)** Policy forbids sending A to Artillery *and* B to Signal. Write this as one linear constraint.

Solution. At most one of the two may be chosen:

$$x_{A,\text{Art}} + x_{B,\text{Sig}} \;\le\; 1.$$

(e) **(5 pts)** Add the constraint and solve both relaxation and IP. Report both solutions and the gap, and explain where the fractions come from.

Solution. Adding `m.addConstr(x[0,1] + x[1,0] <= 1)` and re-solving, continuous then binary:

```
LP: z = 12.5  x = {(0,0): 0.5, (0,1): 0.5, (1,0): 0.5, (1,1): 0.5, (2,2): 1.0}
IP: z = 10.0  x = {(0,0): 1.0, (1,1): 1.0, (2,2): 1.0}
```

Gap $= 12.5 - 10 = 2.5$. The IP sends A $\to$ Signal, B $\to$ Artillery, C $\to$ Engineers.

The new row is not part of the incidence matrix — it puts two 1s in columns that both belong to the *cadet* side of the couple, so condition (iii) can no longer be met and A is no longer TU. The LP exploits exactly that: it splits A and B half-and-half over Signal and Artillery, which costs the forbidden row only $0.5 + 0.5 = 1$ while collecting half of the forbidden pair's fit, $\frac{1}{2}(2+5) + \frac{1}{2}(5+3) + 5 = 12.5$.

Problem 2. Absolute cap or relative cap? **(20 points)**

Two drone teams fly x_1 (North) and x_2 (South) sorties, $x \in \mathbb{Z}^2_+$, to maximize $3x_1 + 2x_2$, with fuel for at most 6 sorties in total. **Version I**: $x_1 \le 3$. **Version II**: $x_1 - x_2 \le 1$.

(a) **(4 pts)** Write both IPs as max $c^\top x$ s.t. $Ax \le b$. Compute every square-submatrix determinant: which is TU? Why can a cap $x_j \le u_j$ never destroy TU?

Solution. $c = (3,2)^\top$ for both, and

$$A_{\mathrm{I}} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad b_{\mathrm{I}} = \begin{pmatrix} 6 \\ 3 \end{pmatrix} \qquad A_{\mathrm{II}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad b_{\mathrm{II}} = \begin{pmatrix} 6 \\ 1 \end{pmatrix}$$

1×1 submatrices are the entries, all $0, \pm 1$ in both. The 2×2:

$$\det A_{\mathrm{I}} = 1 \cdot 0 - 1 \cdot 1 = -1, \qquad \det A_{\mathrm{II}} = 1 \cdot (-1) - 1 \cdot 1 = -2.$$

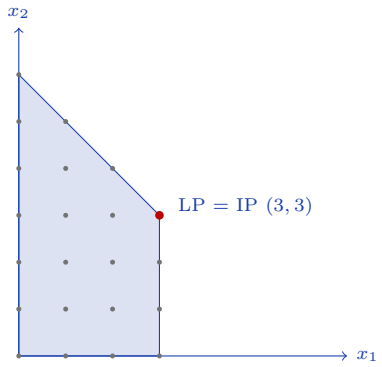
So A_{I} is **TU**, A_{II} is **not**.

A cap $x_j \le u_j$ is a single unit row $e_j^\top$ — one row of I. Appending it keeps A TU: in any square submatrix that row has at most one nonzero, so the determinant is either 0 or ± 1 times a smaller determinant of A. (Equivalently: A TU $\iff \binom{A}{I}$ TU, and deleting rows preserves TU.) A cap restricts *one* variable; only a row that *compares* variables can create a determinant of 2.

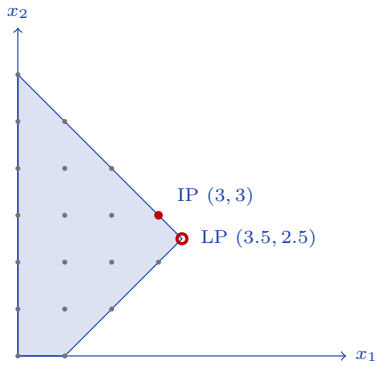
(b) **(6 pts)** Report z_{LP} and z_{IP} for each version and explain the gap using (a).

Solution.

	LP optimum	IP optimum	gap
Version I	$(3,3)$, $z = 15$	$(3,3)$, $z = 15$	0
Version II	$(3.5, 2.5)$, $z = 15.5$	$(3,3)$, $z = 15$	0.5



Version I: $x_1 \le 3$, TU



Version II: $x_1 - x_2 \le 1$, not TU

A_{I} is TU, so with integer b every corner is integral and the LP optimum is the IP optimum — no gap. A_{II} is not TU, so nothing guarantees an integral optimum, and in the figure it is not: the corner where both rows are tight sits at $(3.5, 2.5)$. Hence the gap.

(c) **(5 pts)** Keep A_{II} with integer $b = (b_1, b_2)$, $b_1 \geq b_2 \geq 0$. Solve for the corner where both rows are tight; show it is integral exactly when $b_1 + b_2$ is even. At $b = (7, 1)$ the LP optimum is integral — does that make A_{II} TU?

Solution. $x_1 + x_2 = b_1$ and $x_1 - x_2 = b_2$ give

$$x_1 = \frac{b_1 + b_2}{2}, \qquad x_2 = \frac{b_1 - b_2}{2}.$$

$b_1 + b_2$ and $b_1 - b_2$ have the same parity, so both coordinates are integers exactly when $b_1 + b_2$ is even. At $b = (7, 1)$: $x = (4, 3)$, integral.

No. TU is a property of A alone — it has nothing to do with any particular b. And the implication runs one way only: TU guarantees integral corners, but failing TU never guarantees fractional ones. $b = (7, 1)$ is simply a lucky right-hand side on a non-TU matrix. (That the guarantee really is gone shows at $b = (6, 1)$, which returns $(3.5, 2.5)$.)

(d) **(5 pts)** In Version I, North's real limit is 7 battery packs at 2 per sortie, and a staff officer writes $x_1 \leq 3.5$. Solve the relaxation. Which hypothesis of Hoffman–Kruskal fails? Fix the row by rounding.

Solution. The LP optimum is (3.5, 2.5) with $z = 15.5$ — fractional, even though A_{I} is TU. The failed hypothesis is b **integral**: TU protects you against every *integer* right-hand side, and 3.5 is not one.

Rounding: x_1 is an integer, so $x_1 \leq 3.5 \iff x_1 \leq \lfloor 3.5 \rfloor = 3$. The row becomes $x_1 \leq 3$, b is integral again, and the LP returns $(3, 3)$, $z = 15$ — the IP optimum. Rounding the *bound* is legitimate; rounding the *solution* is what week 2 warned against.

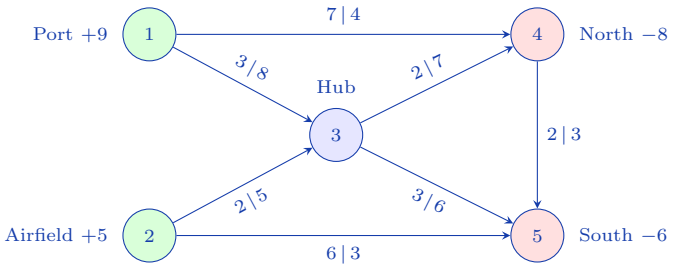
Problem 3. Relief convoys **(20 points)**

Trucks move from two sources through a hub to two divisions, whole truckloads only, at minimum cost; $b = (+9, +5, 0, -8, -6)$ for Port, Airfield, Hub, North, South, with road costs $(3, 7, 2, 6, 2, 3, 2)$ and capacities $(8, 4, 5, 3, 7, 6, 3)$ on $(1, 3), (1, 4), (2, 3), (2, 5), (3, 4), (3, 5), (4, 5)$.

(a) **(3 pts)** Which Week 5 problem models this? Draw the graph and define $G = (V, E)$.

Solution. **Minimum cost network flow (MCNF).**

$$V = \{1, 2, 3, 4, 5\}, \qquad E = \{(1, 3), (1, 4), (2, 3), (2, 5), (3, 4), (3, 5), (4, 5)\}$$



(b) **(4 pts)** Define x; write A, c, b and their dimensions.

Solution. x_a = truckloads on arc a, one per arc: $x \in \mathbb{R}^7$, $c \in \mathbb{R}^7$, $b \in \mathbb{R}^5$, and $A \in \mathbb{R}^{5 \times 7}$ is the node–arc incidence matrix (+1 where the arc leaves, −1 where it arrives). Writing the capacities as rows as well

stacks I underneath:

$$\begin{pmatrix} A \\ I \end{pmatrix} = \begin{array}{l} \\ \text{node 1} \\ \text{node 2} \\ \text{node 3} \\ \text{node 4} \\ \text{node 5} \\ \text{cap (1,3)} \\ \text{cap (1,4)} \\ \text{cap (2,3)} \\ \text{cap (2,5)} \\ \text{cap (3,4)} \\ \text{cap (3,5)} \\ \text{cap (4,5)} \end{array} \begin{array}{c} \begin{array}{ccccccc} (1,3) & (1,4) & (2,3) & (2,5) & (3,4) & (3,5) & (4,5) \end{array} \\ \begin{pmatrix} +1 & +1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & +1 & +1 & 0 & 0 & 0 \\ -1 & 0 & -1 & 0 & +1 & +1 & 0 \\ 0 & -1 & 0 & 0 & -1 & 0 & +1 \\ 0 & 0 & 0 & -1 & 0 & -1 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array} \begin{array}{l} = 9 \\ = 5 \\ = 0 \\ = -8 \\ = -6 \\ \leq 8 \\ \leq 4 \\ \leq 5 \\ \leq 3 \\ \leq 7 \\ \leq 6 \\ \leq 3 \end{array}$$

The top five rows are $Ax = b$, the bottom seven $Ix \leq u$; the objective is $c = (3, 7, 2, 6, 2, 3, 2)^\top$.

(c) **(4 pts)** Explain why A is TU; solve the relaxation and confirm every flow is a whole truckload.

Solution. Checklist: (i) entries are $0, \pm 1$; (ii) every column has exactly two nonzeros, one $+1$ and one -1; (iii) take $M_1 = V$, $M_2 = \emptyset$ — each column sums to $(+1) + (-1) = 0$, and the empty sum is 0. Every node–arc incidence matrix of a directed graph is TU, with no condition on the graph.

```
import gurobipy as gp
from gurobipy import GRB

arcs = [(1,3), (1,4), (2,3), (2,5), (3,4), (3,5), (4,5)]
cost = dict(zip(arcs, [3, 7, 2, 6, 2, 3, 2]))
cap  = dict(zip(arcs, [8, 4, 5, 3, 7, 6, 3]))
b    = {1: 9, 2: 5, 3: 0, 4: -8, 5: -6}
m = gp.Model("convoys")
x = m.addVars(arcs, lb=0, ub=[cap[a] for a in arcs], name="x")   # continuous
m.setObjective(gp.quicksum(cost[a]*x[a] for a in arcs), GRB.MINIMIZE)
for i in b:                                          # outflow - inflow = b_i
    m.addConstr(gp.quicksum(x[a] for a in arcs if a[0] == i)
                - gp.quicksum(x[a] for a in arcs if a[1] == i) == b[i])
m.optimize()
print("z =", m.ObjVal, {a: x[a].X for a in arcs if x[a].X > 1e-6})

z = 73.0 {(1, 3): 8.0, (1, 4): 1.0, (2, 3): 5.0, (3, 4): 7.0, (3, 5): 6.0}
```

Whole truckloads with no integrality declared, as TU promises.

(d) **(5 pts)** Roads $(1,3)$ and $(3,4)$ share a highway carrying at most 10 truckloads. Add it, solve LP and IP, report the gap, and find a 2×2 submatrix with determinant ± 2.

Solution. Adding `m.addConstr(x[1,3] + x[3,4] <= 10)`:

```
LP: z = 78.0 {(1,3): 5.5, (1,4): 3.5, (2,3): 5.0, (3,4): 4.5, (3,5): 6.0}
IP: z = 79.0 {(1,3): 5.0, (1,4): 4.0, (2,3): 5.0, (3,4): 4.0, (3,5): 6.0}
```

Gap $= 79 - 78 = 1$, and the LP ships half truckloads.

The highway row joins the five node rows (capacity rows omitted). Permuting rows and columns changes nothing — TU and determinants survive reordering — so put the highway row next to node 3 and the two shared arcs in the first two columns, and the guilty 2×2 sits in plain sight:

$$\begin{array}{l} \\ \text{node 1} \\ \text{node 2} \\ \text{node 3} \\ \text{highway} \\ \text{node 4} \\ \text{node 5} \end{array} \begin{array}{c} \begin{array}{ccccccc} (1,3) & (3,4) & (1,4) & (2,3) & (2,5) & (3,5) & (4,5) \end{array} \\ \begin{pmatrix} +1 & 0 & +1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & +1 & +1 & 0 & 0 \\ -1 & +1 & 0 & -1 & 0 & +1 & 0 \\ +1 & +1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 & +1 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 \end{pmatrix} \end{array} \begin{array}{l} = 9 \\ = 5 \\ = 0 \\ \leq 10 \\ = -8 \\ = -6 \end{array}$$

The red block is a genuine 2×2 corner of the matrix:

$$\begin{pmatrix} -1 & +1 \\ +1 & +1 \end{pmatrix}, \qquad \det = -1 - 1 = -2.$$

The highway row counts both arcs with $+1$ while node 3 sees one entering and one leaving; that sign clash is the determinant of 2, so the matrix is no longer an incidence matrix and no longer TU.

(e) **(4 pts)** Instead, the Hub receives at most 10: split it into Hub-in and Hub-out joined by a dock road of capacity 10. Still TU? Solve. Lesson?

Solution. The restriction is on the hub's inflow:

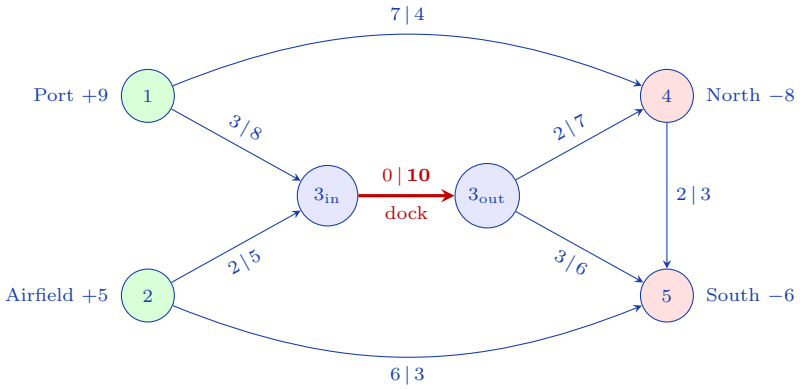
$$x_{13} + x_{23} \;\le\; 10.$$

Written as a row of the matrix it is another (d) — and column $(1,3)$ would then carry three nonzeros, so the checklist no longer applies and the guarantee is gone.

Split node 3 instead. Arcs $(1, 3_{\text{in}})$, $(2, 3_{\text{in}})$ keep their data, arcs $(3_{\text{out}}, 4)$, $(3_{\text{out}}, 5)$ keep theirs, and a dock arc $(3_{\text{in}}, 3_{\text{out}})$ carries everything through at cost 0; both new nodes have $b = 0$. Now *everything* entering the hub must cross the dock, so the same restriction is simply that arc's capacity,

$$0 \;\le\; x_{3_{\text{in}}3_{\text{out}}} \;\le\; 10,$$

a bound on one variable rather than a row across several.



everything entering the hub crosses the dock arc, and that arc is capped at 10

```
LP: z = 76.0 {(1,3in): 8.0, (1,4): 1.0, (2,3in): 2.0, (2,5): 3.0,
              (3in,3out): 10.0, (3out,4): 7.0, (3out,5): 3.0}
IP: z = 76.0   (identical)
```

Still TU — it is again a node–arc incidence matrix, just of a bigger graph — so the LP is integral and the dock arc sits at its capacity, 10.

(The side-row version happens to return 76 integral on this data too. That is 2(c)'s point again: without the structure you have no promise, only luck.)

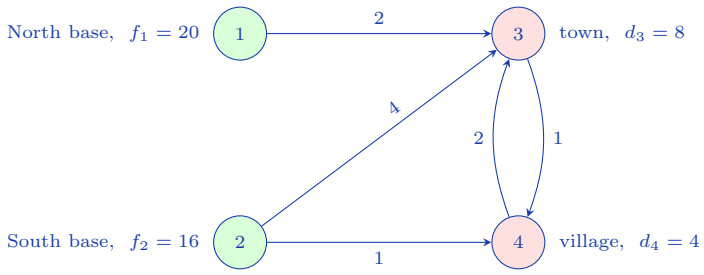
Lesson: as long as every row is still a flow-conservation row, adding nodes and arcs does not destroy TU — the matrix is simply the incidence matrix of a bigger graph — so the LP relaxation is still guaranteed to solve the IP.

Problem 4. The LAP **(20 points)**

Bases 1 (North) and 2 (South) may hold a reserve for town 3 ($d_3 = 8$) and village 4 ($d_4 = 4$); $f_1 = 20$, $f_2 = 16$; roads $c_{13} = 2$, $c_{23} = 4$, $c_{24} = 1$, $c_{34} = 1$, $c_{43} = 2$; $h_i = 1$, $u_i = 40$, $o_i = 1$, $B = 16$, $p = 1$.

(a) **(5 pts)** Write the LAP for this instance with numbers.

Solution.



arc labels: shipping cost per ton; green nodes may open a depot, red nodes have demand

$$
\begin{aligned}
\min\quad & 20x_1 + 16x_2 + r_1 + r_2 + 2y_{13} + 4y_{23} + y_{24} + y_{34} + 2y_{43} + 40(q_3 + q_4) + \textstyle\sum_i s_i \\
\text{s.t.}\quad & -y_{13} + r_1 + q_1 - s_1 = 0 && \text{[node 1]} \\
& -y_{23} - y_{24} + r_2 + q_2 - s_2 = 0 && \text{[node 2]} \\
& y_{13} + y_{23} + y_{43} - y_{34} + q_3 - s_3 = 8 && \text{[node 3]} \\
& y_{24} + y_{34} - y_{43} + q_4 - s_4 = 4 && \text{[node 4]} \\
& r_1 + r_2 \le 16, \qquad r_1 \le 16x_1, \qquad r_2 \le 16x_2, \qquad x_1 + x_2 \le 1 \\
& y, r, s \ge 0, \quad 0 \le q_i \le d_i \ \ (\text{so } q_1 = q_2 = 0), \quad x_1, x_2 \in \{0, 1\}
\end{aligned}
$$

(b) **(5 pts)** Which entries break TU? Show that once x is fixed the rest is TU.

Solution. The linking rows $r_i \le 16x_i$ carry the entry -16 (as $r_i - 16x_i \le 0$). A TU matrix may only contain $0, \pm 1$.

Therefore, with x fixed, $16x_i$ is just an integer bound on r_i, and the remaining constraints are a network: the budget row is the conservation row of one extra *depot* node 0 that feeds r_1 and r_2. Order the free variables as

$$\mathbf{x} \;=\; (\,r_1,\ r_2,\ y_{13},\ y_{23},\ y_{24},\ y_{34},\ y_{43},\ q_3,\ q_4,\ s_1,\ s_2,\ s_3,\ s_4\,)^{\top}$$

($q_1 = q_2 = 0$ because $d_1 = d_2 = 0$). Writing every row as outflow $-$ inflow:

$$
\begin{array}{l}
\\
\text{depot} \\
\text{node 1} \\
\text{node 2} \\
\text{node 3} \\
\text{node 4}
\end{array}
\begin{array}{c}
\begin{array}{cccccccccccccc}
r_1 & r_2 & y_{13} & y_{23} & y_{24} & y_{34} & y_{43} & q_3 & q_4 & s_1 & s_2 & s_3 & s_4
\end{array} \\
\begin{pmatrix}
+1 & +1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & +1 & 0 & 0 & 0 & 0 & 0 & 0 & +1 & 0 & 0 & 0 \\
0 & -1 & 0 & +1 & +1 & 0 & 0 & 0 & 0 & 0 & +1 & 0 & 0 \\
0 & 0 & -1 & -1 & 0 & +1 & -1 & -1 & 0 & 0 & 0 & +1 & 0 \\
0 & 0 & 0 & 0 & -1 & -1 & +1 & 0 & -1 & 0 & 0 & 0 & +1
\end{pmatrix}
\end{array}
\begin{array}{l}
\\
\le 16 \\
= 0 \\
= 0 \\
= -8 \\
= -4
\end{array}
$$

with $0 \le r_i \le 16x_i$, $0 \le q_i \le d_i$, and $y, s \ge 0$.

Every column has at most two nonzeros, and every column that has two carries one $+1$ and one -1 — a node–arc incidence matrix, hence TU. The right-hand side is integer, so the LP relaxation solves the IP.

(c) **(6 pts)** Solve the IP and its relaxation; report both plans and the gap. Then fix x at the IP's value and solve the LP again.

Solution.

```
IP : z = 56.0  x = {1: 0, 2: 1}   r = {2: 12.0}   y = {(2,4): 12.0, (4,3): 8.0}
LP : z = 46.0  x = {1: 0.5, 2: 0.25}  r = {1: 8.0, 2: 4.0}  y = {(1,3): 8.0, (2,4): 4.0}
```

Gap $= 56 - 46 = 10$.

(d) **(4 pts)** Which variables are stage 1 and which stage 2? Keep the IP's stage-1 decisions, let the town's demand be 10 t, and solve stage 2.

Solution. Parts (a)–(c) treated d as known, which in this application it is not: you must choose where to build and what to stock *before* the disaster says where the demand will be. That is why the LAP is read as a **two-stage** problem — commit first, then react once d is realized.

Stage 1 (before the demand is known): x_i, r_i. **Stage 2** (after): y_{ij}, q_i, s_i.

Stage 1 is frozen at $x = (0, 1)$, $r_2 = 12$, costing $16 + 12 = 28$. With $d_3 = 10$ total demand is $14 > 12$, so 2 t must go short:

```
stage 2: z = 108.0  y = {(2,4): 12.0, (4,3): 8.0}  q = {3: 2.0}
total = 28 + 108 = 136
```

The shortage lands at the **town**: both nodes pay the same 40/t penalty, so the model shorts wherever delivery is *dearest* — the town costs $1+2=3$/t to reach through the village, the village 1/t, so shorting the town saves 3 and shorting the village only 1. Recourse $= 12\cdot 1 + 8\cdot 2 + 2\cdot 40 = 108$.

Problem 5. Strong vs weak linking **(20 points)**

Depots D1, D2, D3 and brigades B1, B2, B3 on a ring; each brigade is served by its two neighbours. y_{ij} = fraction of Bj's demand from Di; costs per ton c_{ij} with $c_{11}=1$, $c_{21}=2$, $c_{22}=1$, $c_{32}=2$, $c_{33}=1$, $c_{13}=2$; demands $4, 5, 6$; opening costs $20, 16, 18$.

(a) **(5 pts)** Write the strong and weak formulations. What is the smallest valid M?

Solution. Both share $\min \sum_i f_i x_i + \sum_{ij} c_{ij} d_j y_{ij}$ s.t. $\sum_i y_{ij} = 1$ for every brigade j, $y \geq 0$, $x \in \{0,1\}^3$, and differ only in the linking:

$$\text{(strong)}\ \ y_{ij} \leq x_i \ \text{ for each road } (i,j) \qquad\qquad \text{(weak)}\ \ \sum_j y_{ij} \leq M x_i \ \text{ for each depot } i.$$

Each depot serves exactly two brigades and each $y_{ij} \leq 1$, so $\sum_j y_{ij} \leq 2$ and the smallest valid M is $\mathbf{M = 2}$.

(b) **(4 pts)** Prove the strong formulation is better, with the two-step recipe.

Solution. *Step 1 — $P_{\text{strong}} \subseteq P_{\text{weak}}$.* Sum the two strong rows at depot i: $\sum_j y_{ij} \leq 2x_i = Mx_i$, which is the weak row. So every strong-feasible point is weak-feasible.

Step 2 — the inclusion is strict. Take $x = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, $y_{11} = y_{22} = y_{33} = 1$, all other $y = 0$. Weak holds: $\sum_j y_{1j} = 1 \leq 2 \cdot \frac{1}{2}$. Strong fails: $y_{11} = 1 > x_1 = \frac{1}{2}$. So $P_{\text{strong}} \subsetneq P_{\text{weak}}$, and minimizing over the smaller set can only give a larger (better) lower bound.

(c) **(6 pts)** Solve the IP and both relaxations; verify $z_{\text{LP}}^{\text{weak}} \leq z_{\text{LP}}^{\text{strong}} \leq z_{\text{IP}}$.

Solution.

```
import gurobipy as gp
from gurobipy import GRB

pairs = {1: [1, 2], 2: [2, 3], 3: [3, 1]}        # brigade j -> its two depots
c = {(1,1):1, (2,1):2, (2,2):1, (3,2):2, (3,3):1, (1,3):2}
d = {1: 4, 2: 5, 3: 6};  f = {1: 20, 2: 16, 3: 18}

def solve(kind, binary):
    m = gp.Model(); m.Params.OutputFlag = 0
    vt = GRB.BINARY if binary else GRB.CONTINUOUS
    x = m.addVars([1,2,3], vtype=vt, lb=0, ub=1)
    y = m.addVars(c.keys(), lb=0, ub=1)
    m.setObjective(gp.quicksum(f[i]*x[i] for i in [1,2,3])
                   + gp.quicksum(c[k]*d[k[1]]*y[k] for k in c), GRB.MINIMIZE)
    m.addConstrs(gp.quicksum(y[(i,j)] for i in pairs[j]) == 1 for j in [1,2,3])
    if kind == "strong":
        m.addConstrs(y[k] <= x[k[0]] for k in c)
    else:
        m.addConstrs(gp.quicksum(y[(i,j)] for j in [1,2,3] if (i,j) in c) <= 2*x[i]
                     for i in [1,2,3])
    m.optimize()
    return m.ObjVal, {i: round(x[i].X, 3) for i in [1,2,3]}

for k in ("weak", "strong"):
    print(k, "LP:", solve(k, False))
print("IP    :", solve("strong", True))
```

```
weak LP: (42.0, {1: 0.5, 2: 0.5, 3: 0.5})
strong LP: (49.5, {1: 0.5, 2: 0.5, 3: 0.5})
IP    : (53.0, {1: 0.0, 2: 1.0, 3: 1.0})
```

$42 \le 49.5 \le 53$ as required. The IP opens D2 and D3 and serves B1 from D2.

The **strong** bound is the useful one: it leaves a gap of 3.5 against the weak formulation's 11, on the same integer feasible set. Same answers, a bound three times tighter — and the bound is what branch & bound spends its time on.

(d) **(5 pts)** Derive $x_i + x_{i'} \ge 1$ for each brigade's two depots; which 3×3 matrix is this? Add them, divide by 2 and round up: does the cut close the gap?

Solution. Brigade j needs $y_{ij} + y_{i'j} = 1$ and the strong rows give $y_{ij} \le x_i$, $y_{i'j} \le x_{i'}$, so $x_i + x_{i'} \ge 1$ — one of each brigade's two neighbours must open. The three rows are

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \det = 2,$$

the incidence matrix of the **triangle** — week 5's odd cycle, the non-bipartite case that fails the checklist. Not TU, which is exactly why the LP can sit at $x = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$: every row reads $\frac{1}{2} + \frac{1}{2} = 1$, so adding these three rows alone changes nothing (the strong LP stays at 49.5).

Add them up: $2(x_1 + x_2 + x_3) \ge 3$, divide by 2, and round the right-hand side up because $\sum_i x_i$ is an integer:

$$\textstyle\sum_i x_i \;\ge\; \lceil 1.5 \rceil \;=\; 2.$$

This one cuts off $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, and the strong LP jumps to $z = 53$ with $x = (0, 1, 1)$ — **the gap closes completely**. (On the weak formulation the same cut only reaches 50: a cut helps, but it cannot repair a bad formulation.)